

TeamsHub

August 18, 2026

Multi-Account Workspace & Messaging Hub

1 Project Executive Summary

TeamsHub is an enterprise-grade productivity workspace designed to aggregate multiple Microsoft Teams accounts, guest organization memberships, chats, files, and notifications into one single unified dashboard. Users can seamlessly view, manage, and switch between work, school, client, and guest organizations in 1-click without logging out.

2 Core Problems Solved

- No Repeated Logins** — Eliminates constant login/logout across different Microsoft client tenants.
- Zero Missed Messages** — Consolidates unread notifications and chat alerts from secondary guest organizations.
- No Browser Clutter** — Removes the need to maintain 4–5 separate browser profiles.
- Universal File Access** — Locates shared project files sent across different client organizations in one place.

3 Technology Architecture

Layer	Technology Used	Description
Frontend	React.js, Vite, Context API	Modern Glassmorphic UI with Light/Dark theme support.
Backend API	Node.js, Express.js	Modular REST API server for account & message sync.
Database	MongoDB (Mongoose ORM)	Secure user storage & cached chat metadata.
Authentication	MSAL OAuth 2.0 PKCE	Multi-Tenant Authority (login.microsoftonline.com/common).
Real-Time Engine	Socket.IO + Graph Webhooks	Live message delivery and floating toast notifications.
Mobile App	Android (Kotlin, Jetpack Compose)	MVVM Architecture mobile client.

4 Current Implementation Status

- ✓ **Multi-Tenant MSAL Setup** — Multi-tenant OAuth sign-in (common tenant).
- ✓ **Database & REST APIs** — Mongoose models for Users, Accounts, Chats, Messages.
- ✓ **Real-Time Chat Sync** — Displaying live Teams chat threads and messages.
- ✓ **Socket.IO Real-Time Engine** — Instant message dispatching and toast banners.
- ✓ **Document Preview Modal** — Inline preview for shared PDF, Word (DOCX), and Excel (XLSX) files.
- ✓ **Workspace Switcher** — 1-click account and organization switcher.

5 Testing Deliverables Roadmap^(For Multi-Org Validation)

#	Deliverable	Validation Criteria
01	Multi-Org Sign-In	Test authentication with an email containing multiple tenant memberships.
02	Workspace Switching	Switch between primary and guest organizations in 1-click.
03	Chat Sync & Messaging	Sync teammate chats and test live message dispatching.
04	Real-Time Notifications	Validate audio chime and floating toast banner alerts.
05	Document Hub	Test inline PDF / Word / Excel file preview modal.

6 Future Enhancements

AI Copilot Thread Summarizer Summarize long chat threads using AI models.	Contextual Quick Replies Smart AI reply suggestions for chat messages.	Cross-Tenant Global Search Search documents across all connected client drives simultaneously.
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7 Expected Outcome

A centralized, high-performance platform through which users can manage all their Microsoft Teams accounts, chats, files, and notifications in real-time under one single unified workspace.